Fluid nerve theory: deformation fields, ambient-time filtrations and synthetic differential geometry for a molecular atlas of 71 systems

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with the pronerv fluid-nerve pipeline (code and data provenance in the repository)

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Abstract

The pronerv molecular atlas drew categorical overlap nerves for two systems, alanine dipeptide (heavy atoms) and chignolin CLN025 (Cα sites). This document extends the construction to 71 systems, half atom-based and half backbone-based (35 heavy-atom systems, 36 Cα systems, plus 35 paired heavy-atom representations of the backbone proteins), builds for each a *deformation field* and a *witnessed cover nerve*, runs an *ambient-time filtration* that thickens the cover along the recorded dynamics, and compares the resulting invariants within and between the two groups. Every nerve is also exported as a three-dimensional categorical complex for the proof assistant homotopy.io through Street's orientals. The theoretical part recruits synthetic differential geometry (Kock–Lawvere) to say precisely what a deformation field is at an infinitesimal scale, proves that the nerve of a formally open cover is rigid under every infinitesimal transformation, and hence identifies the measured ambient-time persistence as a finite-time, finite-witness phenomenon whose onset is diagnosed by the mobility exponent. The empirical constructions are descriptive: no equilibrium free energy, validated kinetics or configuration-space nerve theorem is claimed.

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1 Why "fluid" nerves

A nerve is a static object: given a cover $\mathcal{U} = \{U_i\}$ of a space, its simplices record which cover sets meet. Molecular dynamics (MD) data are not static. A recorded trajectory visits the chart, and the overlap pattern that a finite set of frames *witnesses* changes with the time scale at which we look. The earlier atlas treated this by declaring a single cover and a single lag for the structural-mobility height. Here we make the time dependence explicit in two ways:

1. the **deformation field**, a family of conditional statistics of the lag-τ displacement, indexed by the chart position and by the ambient lag τ;
2. the **fluid filtration**, the nested family of nerves of the covers $\{W_i^\tau\}$ obtained by thickening every cover set along the recorded runs by $\pm\tau$ frames.

"Fluid nerve" names the pair (nerve, deformation field) together with its evolution in ambient time. The synthetic differential geometry (SDG) of Section 4 explains what survives of this evolution at an infinitesimal scale, which is less than one might hope and exactly what the data show.

The research questions of the published atlas that this document addresses are: stability of basin and transition descriptions across blocks, temperatures and resolutions (Section 6); an explicit comparison between atomistic and Cα state categories (Section 6.3); and higher-categorical diagramming of the nerves (Section 5).

2 Data and representations

Table 1 lists every system. The **atom-based group** uses all heavy atoms as sites: the mdshare alanine dipeptide (3×250 ns, 300 K, 1 ps frames), twenty capped dipeptides Ace-X-NMe generated here with OpenMM (amber14/ff14SB with GBn2 implicit solvent, 300 K, 1 ps frames; see `generate_dipeptides.py`; these are small new simulations for descriptive geometry, not converged thermodynamics), the MD22 ab initio trajectories (Ac-Ala$_3$-NHMe, docosahexaenoic acid, stachyose, the AT-AT and AT-AT-CG-CG base pairs; DFT MD at 400–500 K, nominal 1 fs frames) and nine MD17 ab initio trajectories of small organic molecules (DFT MD at 500 K, nominal 0.5 fs frames). The **backbone group** uses Cα sites: the CLN025 subset already in the repository (128 adaptive runs, 350 K, physical clock unresolved) and 35 CATH domains from mdCATH (five temperatures 320–450 K × five replicas, frames every 1 ns, CC BY 4.0). Each mdCATH domain is additionally analysed with all heavy atoms on the very same frames; these paired representations are used only for the comparison of Section 6.3.

Coordinates are in nm. Frames are centred and rigidly aligned to an iterated mean structure (Kabsch). The chart is (ϕ, ψ) on the torus for the dipeptide family and the two leading principal components of the aligned site coordinates (fitted on all runs pooled, deterministic sign) otherwise; CLN025 therefore receives a PCA chart here rather than the earlier radius-of-gyration chart. Pair distances use all site pairs, or a fixed random subset of 4000 pairs when a system has more than 100 sites. Provenance (URLs, SHA-256, selection rules, licences) is recorded in every `summary.json`.

Table 1: All primary systems. Objects/inclusions/triangles are the counts of the drawn face-poset nerve; β are GF(2) Betti numbers of the witnessed complex; h is the median supported mobility at the reference lag; α the mobility exponent; r_c the connection radius factor; τ_c the connection time of the fluid filtration.

system	group	frames	sites	objects/incl./tri.	$\beta_{0,1,2}$	r_c	h (nm)	α	τ_c
dipeptide-ala	atom	20,000	10	221/926/780	1,0,0	0.75	0.0428	0.11	0 fs

system	group	frames	sites	objects/incl./tri.	$\beta_{0,1,2}$	r_c	h (nm)	α	τ_c
dipeptide-arg	atom	10,000	16	171/682/564	2,0,0	–	0.0779	0.13	1 ps
dipeptide-asn	atom	20,000	13	256/1108/948	1,0,0	0.75	0.0567	0.10	0 fs
dipeptide-asp	atom	20,000	13	244/1044/888	1,0,0	0.75	0.0561	0.13	0 fs
dipeptide-cys	atom	20,000	11	231/978/828	1,0,0	0.75	0.0402	0.09	0 fs
dipeptide-gln	atom	20,000	14	239/1018/864	1,0,0	0.75	0.0628	0.12	0 fs
dipeptide-glu	atom	20,000	14	208/856/714	1,0,0	0.75	0.065	0.11	0 fs
dipeptide-gly	atom	20,000	9	86/244/162	1,1,0	1.00	0.0457	0.14	0 fs
dipeptide-his	atom	10,000	15	275/1222/1062	1,0,0	0.75	0.0518	0.10	0 fs
dipeptide-ile	atom	10,000	13	349/1650/1482	1,0,0	1.00	0.0362	0.07	0 fs
dipeptide-leu	atom	10,000	13	285/1274/1110	1,0,0	0.75	0.0467	0.12	0 fs
dipeptide-lys	atom	10,000	14	148/568/462	2,0,0	1.25	0.0575	0.15	1 ps
dipeptide-met	atom	10,000	13	236/1008/858	1,0,0	0.75	0.0583	0.14	0 fs
dipeptide-phe	atom	10,000	16	294/1324/1158	1,0,0	0.75	0.0495	0.11	0 fs
dipeptide-pro	atom	10,000	12	258/1152/1008	2,0,0	1.25	0.0201	0.02	1 ps
dipeptide-ser	atom	10,000	11	209/866/726	1,0,0	0.75	0.0411	0.12	0 fs
dipeptide-thr	atom	10,000	12	322/1464/1284	1,0,0	0.75	0.0361	0.08	0 fs
dipeptide-trp	atom	10,000	19	282/1260/1098	1,0,0	0.75	0.0446	0.06	0 fs
dipeptide-tyr	atom	10,000	17	393/1878/1692	1,0,0	0.75	0.042	0.08	0 fs
dipeptide-val	atom	10,000	12	280/1252/1092	1,0,0	0.75	0.0328	0.08	0 fs
sgdml-md17-aspirin	atom	211,762	13	158/592/468	1,0,0	0.75	0.0198	0.54	0 fs
sgdml-md17-azobenzene	atom	99,999	14	63/146/84	1,1,0	0.75	0.0149	0.37	0 fs
sgdml-md17-ethanol	atom	555,092	3	24/12/0	12,0,0	1.50	0.00824	0.39	1.5 fs
sgdml-md17-malonaldehyde	atom	993,237	5	79/210/132	1,0,0	0.75	0.0189	0.52	0 fs
sgdml-md17-naphthalene	atom	326,250	10	41/58/18	2,1,0	1.25	0.00821	0.30	0.5 fs
sgdml-md17-paracetamol	atom	106,490	11	72/188/120	1,1,0	0.75	0.0148	0.36	0 fs
sgdml-md17-salicylic	atom	320,231	10	25/14/0	11,0,0	1.50	0.0097	0.36	11 fs
sgdml-md17-toluene	atom	442,790	7	28/20/0	8,0,0	1.50	0.00765	0.29	1.5 fs
sgdml-md17-uracil	atom	133,770	8	31/34/12	9,0,0	1.25	0.00852	0.35	8.5 fs
sgdml-md22-ac-ala3-nhme	atom	85,109	20	64/140/72	1,6,0	1.00	0.0276	0.62	0 fs
sgdml-md22-at-at	atom	20,001	38	118/412/318	1,0,0	0.75	–	–	0 fs
sgdml-md22-at-at-cg-cg	atom	10,153	76	135/490/384	1,0,0	0.75	–	–	0 fs
sgdml-md22-dha	atom	69,753	24	88/248/162	1,3,0	1.00	0.0339	0.74	0 fs
sgdml-md22-stachyose	atom	27,272	45	144/536/426	1,0,0	0.75	–	0.82	0 fs
ala2	atom	750,000	10	78/212/138	3,3,0	1.25	0.0383	0.07	1 ps
cln025	backbone	63,999	10	84/228/144	1,2,0	1.00	0.197	0.21	0 fr
mdcath-1a92a00-ca	backbone	10,980	50	89/254/168	1,0,0	1.00	0.697	0.21	0 fs
mdcath-1b33n01-ca	backbone	11,090	55	80/216/138	2,3,0	–	0.316	-0.09	1 ns
mdcath-1bdha01-ca	backbone	11,110	57	76/196/120	1,1,0	1.00	0.635	0.05	0 fs
mdcath-1e5wa04-ca	backbone	10,910	55	96/288/198	1,1,0	0.75	0.848	0.21	0 fs
mdcath-1ffsb00-ca	backbone	11,141	68	139/510/402	2,0,0	–	0.419	0.10	1 ns
mdcath-1gjsa00-ca	backbone	10,970	65	86/244/162	1,1,0	0.75	0.604	0.24	0 fs
mdcath-1ha8a00-ca	backbone	11,150	51	78/208/132	1,1,0	1.00	0.21	0.17	0 fs
mdcath-1hcra00-ca	backbone	11,160	52	65/146/78	1,4,0	1.00	0.691	0.23	0 fs
mdcath-1i5xa00-ca	backbone	11,080	61	106/336/240	1,0,0	1.00	0.606	0.21	0 fs
mdcath-1ib8a02-ca	backbone	11,090	67	86/244/162	1,2,0	0.75	0.409	0.06	0 fs
mdcath-1klil00-ca	backbone	11,100	61	80/216/138	1,1,0	1.00	0.552	0.23	0 fs
mdcath-1l6ha00-ca	backbone	11,090	69	77/198/120	1,2,0	1.00	0.344	0.21	0 fs
mdcath-1r8ob01-ca	backbone	11,100	66	87/246/162	1,1,0	0.75	0.908	0.26	0 fs
mdcath-1s6la01-ca	backbone	11,170	52	100/308/216	1,0,0	0.75	0.685	0.24	0 fs
mdcath-1v1hf01-ca	backbone	10,806	73	99/306/216	1,0,0	0.75	0.788	0.22	0 fs
mdcath-1wfwa00-ca	backbone	11,190	74	72/176/102	1,4,0	1.00	0.347	0.13	0 fs
mdcath-1wpbg01-ca	backbone	10,980	54	102/320/228	1,0,0	0.75	0.549	0.22	0 fs
mdcath-1x4pa00-ca	backbone	11,130	66	76/196/120	1,2,0	0.75	0.799	0.19	0 fs
mdcath-1zbqa02-ca	backbone	11,190	58	81/218/138	1,3,0	1.00	0.799	0.19	0 fs
mdcath-2ckka02-ca	backbone	12,500	57	118/392/288	1,1,0	1.00	0.239	0.14	0 fs
mdcath-2d46a00-ca	backbone	11,390	61	90/260/174	1,0,0	0.75	1.02	0.22	0 fs
mdcath-2dmpa01-ca	backbone	12,500	55	76/196/120	1,2,0	1.00	0.886	0.16	0 fs
mdcath-2e3tb03-ca	backbone	11,390	61	116/384/282	1,0,0	1.00	0.489	0.22	0 fs

system	group	frames	sites	objects/incl./tri.	$\beta_{0,1,2}$	r_c	h (nm)	α	τ_c
mdcath-2j8ga03-ca	backbone	12,130	58	91/266/180	1,0,0	0.75	0.732	0.23	0 fs
mdcath-2jrra01-ca	backbone	12,500	53	93/282/198	1,0,0	1.00	0.292	0.14	0 fs
mdcath-2kssa01-ca	backbone	12,500	63	50/92/42	2,2,0	1.25	0.764	0.19	1 ns
mdcath-2lrda00-ca	backbone	12,500	61	74/188/114	1,2,0	1.00	0.21	0.15	0 fs
mdcath-2xepa03-ca	backbone	11,390	53	77/198/120	1,4,0	1.00	0.6	0.22	0 fs
mdcath-2yila02-ca	backbone	12,500	51	96/300/216	1,0,0	1.00	0.247	0.17	0 fs
mdcath-3g5ca02-ca	backbone	12,500	55	85/238/156	1,0,0	1.00	0.476	0.22	0 fs
mdcath-3pmqa02-ca	backbone	12,130	50	68/160/90	1,4,0	0.75	0.872	0.17	0 fs
mdcath-4a0ta01-ca	backbone	11,390	61	75/190/114	1,3,0	1.00	0.688	0.14	0 fs
mdcath-4dcza00-ca	backbone	11,020	55	81/218/138	1,1,0	1.00	0.777	0.22	0 fs
mdcath-5b1aj00-ca	backbone	11,950	58	76/196/120	1,3,0	1.00	0.58	0.23	0 fs
mdcath-5daea00-ca	backbone	11,390	57	72/172/96	1,5,0	1.00	0.281	0.11	0 fs

3 Constructions

3.1 Witnessed cover nerve

Let $q_t \in C$ be the chart coordinate of frame t and S the finite set of all frames. Eighteen centres are chosen by farthest-point traversal among occupied grid cells (30×30 bins), and the cover $\mathcal{U} = \{U_i\}$ consists of balls of a fixed radius in the declared chart metric (55° on the torus; 0.8 standard deviations per axis otherwise), exactly as in the measured revision of the atlas. The membership matrix $M_{ti} = [q_t \in U_i]$ gives the *witnessed complex*

$$K(\mathcal{U}; S) = \{\sigma \subseteq \{1, \dots, 18\} : \ \exists t \in S \ \ \forall i \in \sigma, \ q_t \in U_i\},$$

stored by maximal simplices without dimension truncation. The drawing of the atlas is the nerve of the face poset of the 2-skeleton: objects are faces of dimension ≤ 2, morphisms are strict inclusions, and composable pairs are its triangles. We report GF(2) Betti numbers $\beta_0, \beta_1, \beta_2$ of K, counts of simplices by dimension, the maximal dimension, and a radius sensitivity ladder (0.75, 1, 1.25, 1.5 times the base radius). Candidate centres are occupied cells with at least $\max(10, \lfloor |S|/30000 \rceil)$ frames (25 for alanine, as in the atlas). Farthest-point traversal deliberately reaches the fringe of the occupied chart; for a unimodal vibrational chart (small MD17 molecules) several fringe balls contain only their own few frames and stay isolated at the base radius, so β_0 at the base radius counts isolated witnessed balls as well as genuine components. The *connection radius factor*, the smallest ladder multiple at which $\beta_0 = 1$, is reported alongside. For alanine dipeptide this reproduces the published measured atlas exactly (same centres, 17 maximal simplices, 78 objects, 212 inclusions, 138 triangles, $\beta = (3, 3, 0)$, 113 uncovered frames).

3.2 Deformation field

Fix a lag τ in frames. For every within-run pair $(t, t+\tau)$ we record

$$h_\tau(t) = \Big(\tfrac{1}{P} \sum_p \big(d_p(x_{t+\tau}) - d_p(x_t)\big)^2\Big)^{1/2} \quad \text{(mobility, nm; } d_p = \text{site pair distances)}$$

$$\Delta q_\tau(t) = q_{t+\tau} - q_t \quad \text{(chart displacement; periodic difference on the torus)}$$

$$u_\tau(t) = R_t\, x_{t+\tau} - x_t \quad \text{(site displacement after Kabsch alignment } R_t \text{ of frame } t+\tau \text{ onto } t)$$

and bin them by q_t. The **deformation field at scale** τ is the collection of conditional means and covariances per bin: $h_\tau(b)$, $v_\tau(b) = \mathbb{E}[\Delta q_\tau \mid b]$, $\Sigma_\tau(b) = \mathrm{Cov}[\Delta q_\tau \mid b]$ and $u_\tau(b) \in (\mathbb{R}^3)^K$. Pairs are kept only inside contiguous blocks (about ten per run, never across run boundaries); a bin is supported when it has at least 100 pairs, 5 blocks and 2 runs. The field grid has

$\min(30, \max(10, \lfloor\sqrt{|S|/150}\rfloor))$ bins per axis so that the support rule is attainable for small deposits (10×10 for an 11 000-frame mdCATH domain, 30×30 for alanine); the cover candidate grid and the flooding grid stay at 30×30. Block resampling (200 draws) gives a conditional spread for h, which is not a calibrated confidence interval. The lag ladder is geometric; the atlas height uses the reference lag of each family (100 ps for the dipeptides, 10 frames for CLN025 and mdCATH, 100 fs for MD22, 100 fs for MD17). The **mobility exponent** is the log–log slope of the median supported h_τ against τ (physical time when a clock exists).

3.3 Fluid filtration (ambient time)

For $\tau \geq 0$ let $W_i^\tau = \{t \in S : \exists s$ in the same run, $|s-t| \leq \tau,\ q_s \in U_i\}$: the frames within ambient time τ of a visit to U_i. Since $W_i^0 = \{t : q_t \in U_i\}$ and $W_i^\tau \subseteq W_i^{\tau'}$ for $\tau \leq \tau'$, the witnessed nerves $N_\tau = K(\{W_i^\tau\}; S)$ form a filtration of simplicial complexes starting at the ordinary overlap nerve. We compute Betti curves $\tau \mapsto \beta_k(N_\tau)$ over the lag ladder and the persistence pairs (birth τ, death τ) over GF(2) by column reduction. The *connection time* τ_c is the first ladder value at which $\beta_0(N_\tau) = 1$. Nothing here is a rate: τ_c is the ambient time at which the recorded runs tie every cover set to every other through actual visits.

3.4 Flooding filtration (the metadynamics analogue)

Let $F = -\log(p/p_{\max})$ be the smoothed negative log occupancy on the chart grid (never-visited bins are never flooded). The sublevel sets $\{F \leq c\}$ form a cubical filtration (periodic on the torus); we compute its Betti curves and the exact 0-dimensional persistence pairs (basin birth level, merge level) by union–find with the elder rule. In metadynamics a history-dependent bias fills basins in the order of their depth, and in the ideal limit the bias converges to $-F$; the flooding filtration is thus the geometry that a metadynamics run of this chart would trace out, computed from the recorded occupancy without any biased simulation. The merge levels are reported in units of log-occupancy, not free energy.

4 Synthetic differential geometry of deformation fields

We now say what a deformation field is at an infinitesimal scale and what that implies for nerves. The setting is a well-adapted model of SDG (a smooth topos $\mathcal{E}$ containing the category of manifolds) with the Kock–Lawvere axiom: for the line R and $\mathbb{D} = \{d \in R : d^2 = 0\}$, every map $f : \mathbb{D} \to R$ is uniquely of the form $f(d) = f(0) + d\,f'(0)$ [1, 3]. Objects such as the shape space M of K sites modulo rigid motions are microlinear, and $M^{\mathbb{D}}$ is the tangent bundle.

Definition 4.1 (infinitesimal deformation field)**.** *A* deformation field *on M is a vector field in the synthetic sense: a map $X : M \times \mathbb{D} \to M$ with $X(x, 0) = x$, equivalently a family of* infinitesimal transformations *$X_d = X(-, d) : M \to M$, $d \in \mathbb{D}$.*

Proposition 4.2 (infinitesimal transformations are invertible)**.** *For a vector field X on a microlinear object, X_d is a bijection with inverse X_{-d} for every $d \in \mathbb{D}$.*

Proof. For $(d_1, d_2) \in \mathbb{D}(2) = \{d_1^2 = d_2^2 = d_1 d_2 = 0\}$ microlinearity gives the flow law $X(X(x, d_1), d_2) = X(x, d_1 + d_2)$ (Kock [1, I.8]). Since $(d, -d) \in \mathbb{D}(2)$, $X_{-d} \circ X_d = X_0 = \mathrm{id}$ and symmetrically. □

Definition 4.3 (first-order mobility)**.** *For an observable $f : M \to R$ (for us a pair distance d_p) and a deformation field X, the Kock–Lawvere axiom applied to $d \mapsto f(X_d(x))$ gives a unique $(\mathcal{L}_X f)(x)$ with*

$$f(X_d(x)) = f(x) + d\,(\mathcal{L}_X f)(x) \qquad (d \in \mathbb{D}),$$

the Lie derivative. The first-order mobility *of X at x is $\|\mathcal{L}_X d\|_{\mathrm{rms}}(x) = \big(\frac{1}{P}\sum_p (\mathcal{L}_X d_p)^2(x)\big)^{1/2}$.*

Remark 4.4 (what the measured h_τ estimates)**.** *If the recorded dynamics were the flow of a classical vector field X sampled at an interval τ small compared with the curvature of the trajectories, then $h_\tau(t) = \tau\,\|\mathcal{L}_X d\|_{\mathrm{rms}}(x_t) + O(\tau^2)$, so h_τ/τ estimates the first-order mobility and the mobility exponent is* 1 *(ballistic regime). If the dynamics is diffusive at the resolved scale, the appropriate synthetic object is a second-order one (a map on $\mathbb{D}_2 = \{d^3 = 0\}$ or an Itô-type calculus) and $h_\tau \propto \tau^{1/2}$. If τ exceeds the decorrelation time of the internal geometry inside a chart bin, h_τ saturates at the conditional diameter and the exponent is* 0*. Thus the mobility exponent, which we measure for every system, tells which synthetic description applies at the resolved sampling interval. The chart bin is a fibre; the conditional expectation over the fibre is unavoidable and is why the field is a* conditional *object.*

Theorem 4.5 (rigidity of nerves under infinitesimal transformations)**.** *Let $\mathcal{U} = \{U_i\}$ be a finite cover of M by Penon-open subobjects and X a vector field. For every $d \in \mathbb{D}$ and every i, $X_d(U_i) = U_i$. Consequently the thickened cover $U_i^d := U_i \cup X_d(U_i) \cup X_{-d}(U_i)$ equals $\mathcal{U}$ and its nerve is unchanged: $\mathcal{N}(\mathcal{U}^d) = \mathcal{N}(\mathcal{U})$.*

Proof. Penon-open subobjects are infinitesimally stable: if $x \in U$ and y is a first-order neighbour of x (in a chart, $y - x \in \mathbb{D}(n)$), then $y \in U$ [4, 1]. For a vector field, $X_d(x)$ is a first-order neighbour of x, so $X_d(U_i) \subseteq U_i$; applying the same to X_{-d} and Proposition 4.2 gives equality. □

Remark 4.6 (what this says about the fluid filtration)**.** *Theorem 4.5 is a no-go result for a naive "infinitesimal fluid nerve": in the smooth, open, infinitely sampled idealisation, the ambient-time filtration has no first-order term at $\tau = 0$; nerve changes are finite-time effects. On recorded data two things break the idealisation. First, the witness set is finite, so N_τ is the nerve of a cover of a finite set, whose combinatorics changes as soon as a single new frame enters a new intersection. Second, the saved frames are rarely infinitesimally close: at 1 ps (dipeptides) or 1 ns (mdCATH) the internal geometry has already decorrelated (saturated exponent), while at 0.5–1 fs (MD17/MD22) consecutive frames are genuinely close (ballistic exponent). The theorem therefore predicts, and the measurements can test, that the initial part of the Betti curves is flat when and only when the mobility exponent is close to one.*

Proposition 4.7 (the deformation field is a cosheaf of measures on the cover)**.** *Let μ_τ be the joint law of $(x_t, x_{t+\tau})$ on $M \times M$ (empirically, the counting measure on recorded pairs). The assignment $U \mapsto \mu_\tau|_{U\times M}$ on the face poset of $\mathcal{N}(\mathcal{U})$ (with $U_\sigma = \bigcap_{i\in\sigma} U_i$ and inclusions $U_{\sigma'} \subseteq U_\sigma$ for $\sigma \subseteq \sigma'$) is a cosheaf: for finite unions, $\mu|_{U\cup V} = \mu|_U + \mu|_V - \mu|_{U\cap V}$. The derived statistics $v_\tau, \Sigma_\tau, h_\tau, u_\tau$ are ratios of such measures and form only a presheaf of conditional laws (they restrict, but do not glue additively).*

Proof. Restriction of a measure to measurable subsets satisfies inclusion–exclusion; conditional means are ratios and are not additive. □

Definition 4.8 (fluid nerve)**.** *The* fluid nerve *of a system at scale τ is the triple $(\mathcal{N}(\mathcal{U}), \mu_\tau|_{(-)}, \tau \mapsto N_\tau)$: the face-poset nerve, the cosheaf of lag-τ pair measures over it (from which the deformation field is derived), and the ambient-time filtration of thickened nerves. Its invariants used below are the counts of the drawn nerve, the Betti numbers of K, the per-bin field at the reference lag, the mobility exponent, the connection time and the β_1 persistence, and the flooding basin structure.*

Remark 4.9 (Kock's infinitesimal simplices and the cover nerve)**.** *Kock's combinatorial differential forms live on the simplicial object of infinitesimal simplices $(x_0, \ldots, x_k)$ of mutually first-order neighbours [2]. The Čech nerve of the cover of M by first-order monads $\mathfrak{M}(x) = \{y : y \sim x\}$ has as 1-simplices the pairs (x, x') with a common neighbour, i.e.* second-*order neighbours. Thickening a cover by a flow, as in Section 3.3, composes neighbour relations in the same way: N_τ records chains of visits of total length $\leq 2\tau$ around an overlap. The finite-scale ladder of N_τ is thus the data analogue of the tower of k-th order neighbour complexes; the rigidity theorem says the tower starts flat.*

Conjecture 4.10 (finite-time fluidity)**.** *For an ergodic smooth flow with bounded speed s on a compact M and a cover by balls of radius r in a Lipschitz chart, the connection time of the fluid filtration of the idealised (infinite-witness) cover satisfies $\tau_c \leq \mathrm{diam}(\mathit{chart})/s$, and $\beta_1(N_\tau)$ is eventually zero. The empirical connection times of Section 6 are finite-sample lower bounds for this ideal quantity; we do not prove the conjecture here.*

5 Three-dimensional categorical complexes in homotopy.io

Street's orientals turn an ordered simplicial complex into a free ω-category. We use them up to dimension three and present each nerve to homotopy.io [6]: 0-cells are vertices, 1-cells $i \to j$ $(i < j)$ are edges, a triangle $[i, j, k]$ is a 2-cell $(ij; jk) \Rightarrow ik$, and a tetrahedron $[i, j, k, l]$ is a 3-cell from the pasting "$[i, j, k]$ then $[i, k, l]$" to the pasting "$[j, k, l]$ then $[i, j, l]$", both running from $i \to j \to k \to l$ to $i \to l$. Two encodings are exported for every system: the witnessed cover complex itself (at most 18 objects) and the order complex of its face poset restricted to faces of dimension ≤ 3 (the categorical object the atlas draws, with length-three chains as 3-cells). The files are produced by replaying the construction through the tool's own proof model (its globularity and deep diagram checks are run on every cell) and serialising with its native format; the companion document `docs/homotopy-io-instructions.md` gives the import steps, the manual click script and how to read the 3D rendering. Table 2 lists the exported signature sizes. No cell is marked invertible: witnessing an overlap is a directed fact, and nothing in the presentation asserts that overlaps compose dynamically.

Table 2: homotopy.io exports (`dist/homotopy/<system>/`): generator counts of the base encoding by dimension and of the face-poset encoding; status after the tool's deep diagram check.

system	base (0/1/2/3-cells)	status	poset generators	status
ala2	18/37/23/4	ok	728	ok
cln025	18/42/24/1	ok	531	ok
dipeptide-ala	18/73/130/120	ok	10927	skipped
dipeptide-arg	18/59/94/81	ok	7492	skipped
dipeptide-asn	18/80/158/170	ok	15062	skipped
dipeptide-asp	18/78/148/143	ok	12901	skipped
dipeptide-cys	18/75/138/129	ok	11712	skipped
dipeptide-gln	18/77/144/138	ok	12471	skipped
dipeptide-glu	18/71/119/100	ok	9278	skipped
dipeptide-gly	18/41/27/4	ok	792	ok
dipeptide-his	18/80/177/237	ok	20334	skipped
dipeptide-ile	18/84/247/503	ok	41206	skipped
dipeptide-leu	18/82/185/252	ok	21569	skipped
dipeptide-lys	18/53/77/63	ok	5903	skipped
dipeptide-met	18/75/143/151	ok	13427	skipped
dipeptide-phe	18/83/193/290	ok	24526	skipped
dipeptide-pro	18/72/168/252	ok	21318	skipped
dipeptide-ser	18/70/121/113	ok	10276	skipped
dipeptide-thr	18/90/214/287	ok	24595	skipped
dipeptide-trp	18/81/183/250	ok	21390	skipped
dipeptide-tyr	18/93/282/578	ok	47313	skipped
dipeptide-val	18/80/182/266	ok	22574	skipped
mdcath-1a92a00-ca	18/43/28/2	ok	661	ok
mdcath-1a92a00-heavy	18/47/37/8	ok	1240	ok
mdcath-1b33n01-ca	18/39/23/3	ok	659	ok
mdcath-1b33n01-heavy	18/36/14/1	ok	383	ok
mdcath-1bdha01-ca	18/38/20/0	ok	392	ok
mdcath-1bdha01-heavy	18/38/21/0	ok	405	ok
mdcath-1e5wa04-ca	18/45/33/6	ok	1032	ok

system	base (0/1/2/3-cells)	status	poset generators	status
mdcath-1e5wa04-heavy	18/45/33/5	ok	957	ok
mdcath-1ffsb00-ca	18/54/67/37	ok	3826	ok
mdcath-1ffsb00-heavy	18/51/59/30	ok	3188	ok
mdcath-1gjsa00-ca	18/41/27/4	ok	792	ok
mdcath-1gjsa00-heavy	18/46/35/7	ok	1136	ok
mdcath-1ha8a00-ca	18/38/22/2	ok	568	ok
mdcath-1ha8a00-heavy	18/39/21/2	ok	558	ok
mdcath-1hcra00-ca	18/34/13/0	ok	289	ok
mdcath-1hcra00-heavy	18/38/19/2	ok	529	ok
mdcath-1i5xa00-ca	18/48/40/9	ok	1357	ok
mdcath-1i5xa00-heavy	18/44/32/8	ok	1166	ok
mdcath-1ib8a02-ca	18/41/27/5	ok	867	ok
mdcath-1ib8a02-heavy	18/43/35/10	ok	1352	ok
mdcath-1klil00-ca	18/39/23/2	ok	584	ok
mdcath-1klil00-heavy	18/38/21/3	ok	630	ok
mdcath-1l6ha00-ca	18/39/20/0	ok	395	ok
mdcath-1l6ha00-heavy	18/44/32/6	ok	1016	ok
mdcath-1r8ob01-ca	18/42/27/3	ok	720	ok
mdcath-1r8ob01-heavy	18/42/25/1	ok	544	ok
mdcath-1s6la01-ca	18/46/36/7	ok	1149	ok
mdcath-1s6la01-heavy	18/51/49/16	ok	2008	ok
mdcath-1v1hf01-ca	18/45/36/8	ok	1221	ok
mdcath-1v1hf01-heavy	18/43/29/4	ok	824	ok
mdcath-1wfwa00-ca	18/37/17/1	ok	425	ok
mdcath-1wfwa00-heavy	18/39/23/2	ok	584	ok
mdcath-1wpbg01-ca	18/46/38/10	ok	1400	ok
mdcath-1wpbg01-heavy	18/46/38/10	ok	1400	ok
mdcath-1x4pa00-ca	18/38/20/1	ok	467	ok
mdcath-1x4pa00-heavy	18/39/19/2	ok	532	ok
mdcath-1zbqa02-ca	18/40/23/3	ok	662	ok
mdcath-1zbqa02-heavy	18/34/13/0	ok	289	ok
mdcath-2ckka02-ca	18/52/48/16	ok	1998	ok
mdcath-2ckka02-heavy	18/64/72/27	ok	3171	ok
mdcath-2d46a00-ca	18/43/29/3	ok	749	ok
mdcath-2d46a00-heavy	18/44/30/3	ok	765	ok
mdcath-2dmpa01-ca	18/38/20/1	ok	467	ok
mdcath-2dmpa01-heavy	18/45/32/4	ok	869	ok
mdcath-2e3tb03-ca	18/51/47/13	ok	1757	ok
mdcath-2e3tb03-heavy	18/56/69/38	ok	3933	ok
mdcath-2j8ga03-ca	18/43/30/4	ok	837	ok
mdcath-2j8ga03-heavy	18/35/17/2	ok	494	ok
mdcath-2jrra01-ca	18/42/33/8	ok	1173	ok
mdcath-2jrra01-heavy	18/53/52/19	ok	2278	ok
mdcath-2kssa01-ca	18/25/7/0	ok	184	ok
mdcath-2kssa01-heavy	18/31/14/1	ok	368	ok
mdcath-2lrda00-ca	18/37/19/1	ok	451	ok
mdcath-2lrda00-heavy	18/47/40/11	ok	1504	ok
mdcath-2xepa03-ca	18/39/20/2	ok	545	ok
mdcath-2xepa03-heavy	18/41/25/4	ok	766	ok
mdcath-2yila02-ca	18/42/36/13	ok	1587	ok
mdcath-2yila02-heavy	18/35/20/2	ok	533	ok
mdcath-3g5ca02-ca	18/41/26/2	ok	629	ok
mdcath-3g5ca02-heavy	18/39/21/1	ok	483	ok
mdcath-3pmqa02-ca	18/35/15/1	ok	393	ok
mdcath-3pmqa02-heavy	18/37/17/1	ok	425	ok
mdcath-4a0ta01-ca	18/38/19/1	ok	454	ok
mdcath-4a0ta01-heavy	18/43/30/5	ok	912	ok
mdcath-4dcza00-ca	18/40/23/1	ok	512	ok

system	base (0/1/2/3-cells)	status	poset generators	status
mdcath-4dcza00-heavy	18/40/26/3	ok	701	ok
mdcath-5b1aj00-ca	18/38/20/2	ok	542	ok
mdcath-5b1aj00-heavy	18/41/27/4	ok	792	ok
mdcath-5daea00-ca	18/38/16/0	ok	340	ok
mdcath-5daea00-heavy	18/39/23/3	ok	659	ok
sgdml-md17-aspirin	18/62/78/40	ok	4218	skipped
sgdml-md17-azobenzene	18/31/14/1	ok	368	ok
sgdml-md17-ethanol	18/6/0/0	ok	36	ok
sgdml-md17-malonaldehyde	18/39/22/0	ok	421	ok
sgdml-md17-naphthalene	18/20/3/0	ok	117	ok
sgdml-md17-paracetamol	18/34/20/4	ok	680	ok
sgdml-md17-salicylic	18/7/0/0	ok	39	ok
sgdml-md17-toluene	18/10/0/0	ok	48	ok
sgdml-md17-uracil	18/11/2/0	ok	77	ok
sgdml-md22-ac-ala3-nhme	18/34/12/1	ok	351	ok
sgdml-md22-at-at	18/47/53/31	ok	3173	ok
sgdml-md22-at-at-cg-cg	18/53/64/37	ok	3784	ok
sgdml-md22-dha	18/43/27/4	ok	798	ok
sgdml-md22-stachyose	18/55/71/46	ok	4556	skipped
106 systems exported; 85 with both encodings accepted by the deep check.				

6 Results

All numbers are produced by `compare_all.py` from the per-system summaries; the tables below are generated, not transcribed.

6.1 Within and between groups

invariant	median atom	median backbone	Mann–Whitney p	Cliff's δ	n
nerve objects	171	82.5	0.002	+0.43	35/36
inclusions	682	223	0.002	+0.44	35/36
triangles	564	141	0.001	+0.45	35/36
max simplex dim	6	3	0.000	+0.50	35/36
$\beta 0$ (cover)	1	1	0.040	+0.19	35/36
$\beta 1$ (cover)	0	1	0.000	-0.48	35/36
median h at τ_ref (nm)	0.0393	0.602	0.000	-1.00	32/36
drift/spread at τ_ref	0.52	0.425	0.001	+0.47	32/36
mobility exponent	0.129	0.206	0.397	-0.12	33/36
τ_connect (frames)	0	0	0.042	+0.18	35/36
$\beta 1$ total lifetime (frames)	0	2	0.089	-0.22	35/36
flooding basins (persistence ≥ 1)	4	18.5	0.000	-0.91	35/36
occupancy entropy (bits)	6.64	7.45	0.225	-0.17	35/36

Table 3: Group medians and descriptive two-sided tests (atom-based vs backbone-based). Mean within-group distance atom 6.05, backbone 3.26, between 5.95.

Figure 1 embeds the primary systems by their z-scored invariants (classical MDS) and Figure 2 shows the group distributions of six of them. The between/within distance ratio is 1.29 (permutation $p = 0.000$, 2000 label permutations). Figure 3 shows the mobility ladders and Figure 4 the fluid-filtration component curves normalised by each system's connection time.

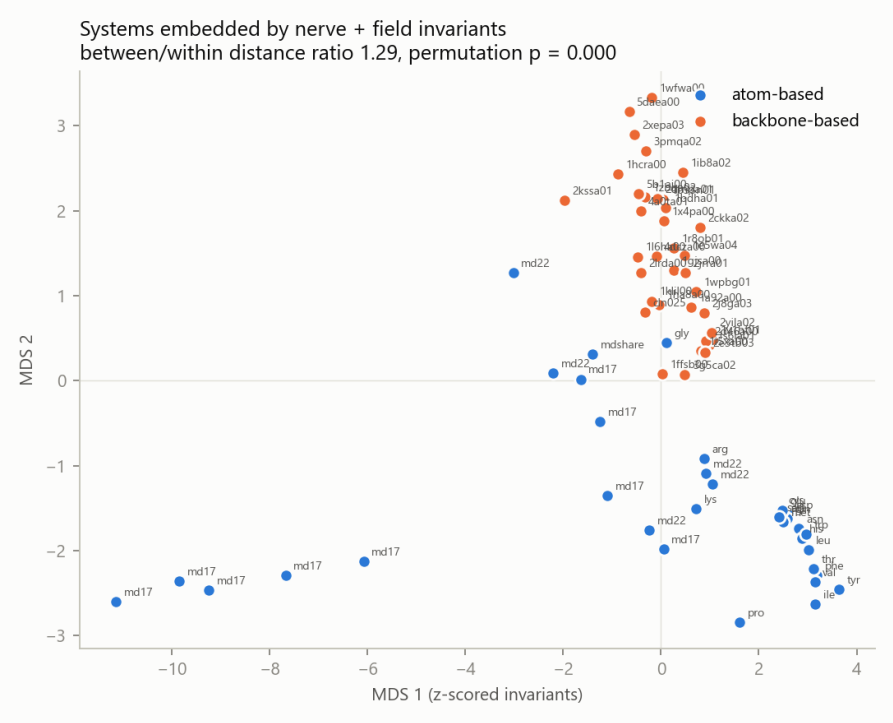


Figure 1: Systems embedded by nerve and deformation-field invariants. Blue: atom-based; orange: backbone-based.

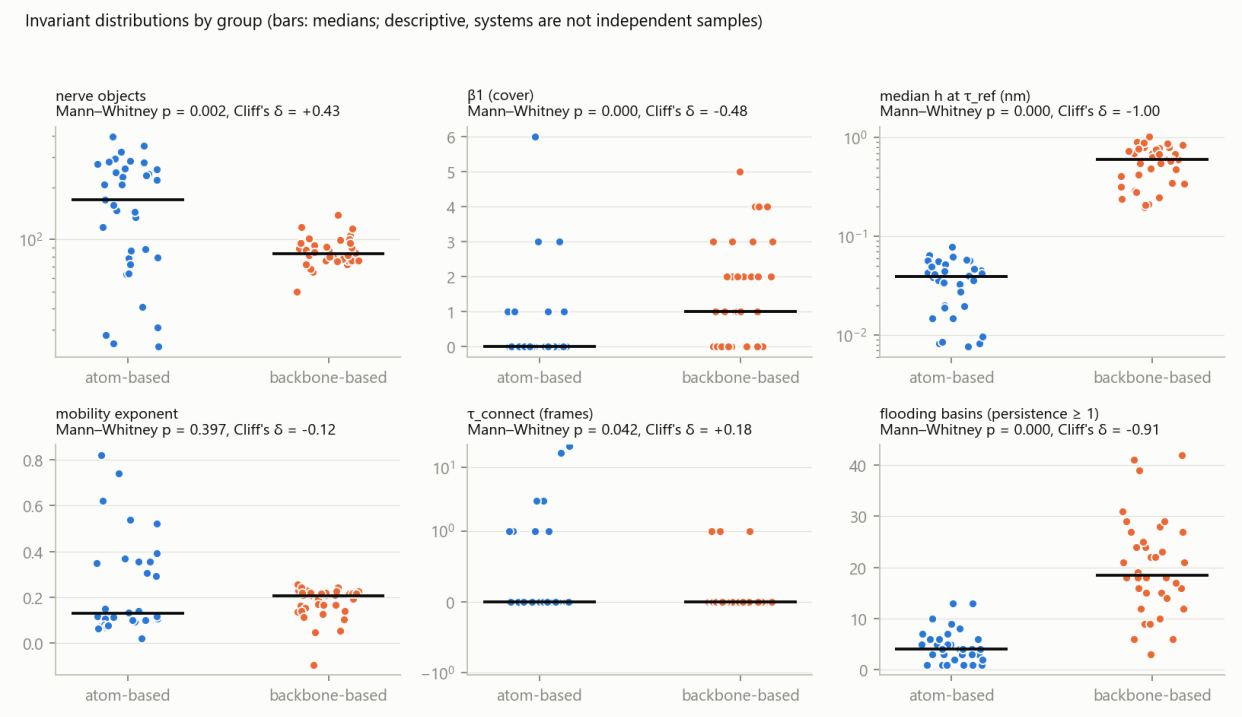


Figure 2: Invariant distributions by group. Bars mark medians; tests are descriptive because systems are not independent samples (the dipeptide family shares a protocol, the mdCATH domains share a dataset).

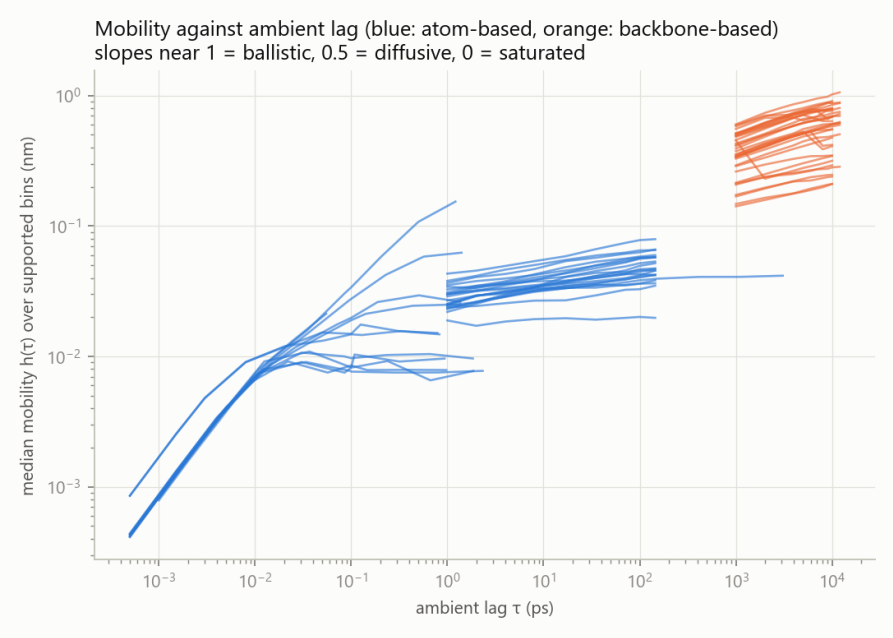


Figure 3: Median mobility against ambient lag. Ab initio trajectories sampled every femtosecond show the ballistic regime; picosecond- and nanosecond-sampled systems are saturated.

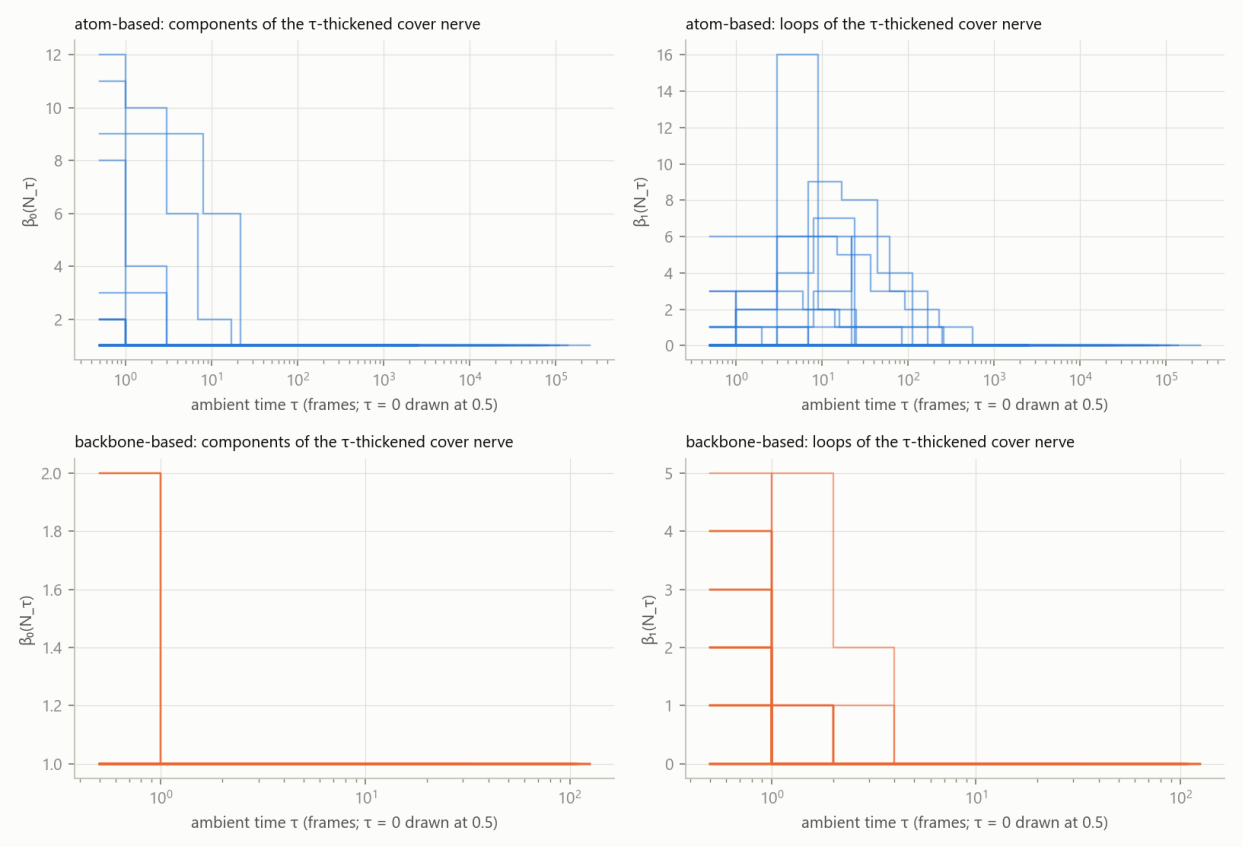


Figure 4: Fluid filtration: $\beta_0(N_\tau)/\beta_0(N_0)$ against τ/τ_c.

6.2 Rigidity test

Theorem 4.5 predicts that cover membership is invariant between frames that are first-order close relative to the cover. We measure the *membership flip rate*: the fraction of consecutive within-run

frame pairs whose 18-ball membership vector differs. Classifying by the early mobility exponent α_{early} (log–log slope over the first half of the lag ladder), 12 of the 69 primary systems are ballistic ($\alpha_{\text{early}} > 0.6$) and 32 are saturated ($\alpha_{\text{early}} < 0.2$). The mean one-frame flip rate is 0.023 for the ballistic systems and 0.782 for the saturated ones (Spearman $\rho = -0.76$, $p = 0.000$, $n = 69$ between the early exponent and the flip rate). The relative drop of β_0 between $\tau = 0$ and the first ladder lag is 0.15 (ballistic) against 0.08 (saturated); this statistic is dominated by isolated fringe balls, which a single frame of thickening attaches to the main component, so the flip rate is the cleaner test. Ballistic systems: sgdml-md17-aspirin, sgdml-md17-azobenzene, sgdml-md17-ethanol, sgdml-md17-malonaldehyde, sgdml-md17-naphthalene, sgdml-md17-paracetamol, sgdml-md17-salicylic, sgdml-md17-toluene, sgdml-md17-uracil, sgdml-md22-ac-ala3-nhme, sgdml-md22-dha, sgdml-md22-stachyose.

6.3 Comparison between atomistic and Cα state categories

For each mdCATH domain the heavy-atom and Cα covers are built on identical frames. Mapping every heavy-atom face σ to the Cα face with the largest witness-set Jaccard index, $f(\sigma)$, we ask whether strict inclusions and composable pairs are preserved (laxly, allowing equal images). Table 4 and Figure 5 report the rates. A rate of one in both directions would make f a functor between the face posets; the measured shortfall quantifies the approximation error asked for in the atlas's sixth research question.

domain	faces (heavy/Cα)	med. Jaccard	incl. lax	incl. strict	comp. lax	Cα →heavy incl.
1a92a00	110/91	0.17	0.62	0.36	0.42	0.69
1b33n01	69/83	0.10	0.44	0.25	0.16	0.48
1bdha01	77/76	0.28	0.75	0.54	0.59	0.72
1e5wa04	101/102	0.28	0.63	0.42	0.46	0.69
1ffsb00	158/176	0.21	0.53	0.36	0.31	0.63
1gjsa00	106/90	0.26	0.72	0.48	0.54	0.60
1ha8a00	80/80	0.20	0.59	0.37	0.33	0.59
1hcra00	77/65	0.21	0.74	0.51	0.64	0.64
1i5xa00	102/115	0.28	0.67	0.53	0.43	0.81
1ib8a02	106/91	0.15	0.64	0.42	0.45	0.49
1klil00	80/82	0.17	0.64	0.39	0.44	0.64
1l6ha00	100/77	0.21	0.70	0.45	0.52	0.67
1r8ob01	86/90	0.30	0.73	0.58	0.56	0.69
1s6la01	134/107	0.26	0.62	0.43	0.40	0.72
1v1hf01	94/107	0.30	0.72	0.54	0.58	0.66
1wfwa00	82/73	0.28	0.72	0.51	0.57	0.74
1wpbg01	112/112	0.27	0.73	0.51	0.53	0.62
1x4pa00	78/77	0.20	0.66	0.44	0.41	0.67
1zbqa02	65/84	0.21	0.72	0.49	0.54	0.69
2ckka02	181/134	0.14	0.63	0.40	0.46	0.56
2d46a00	95/93	0.29	0.56	0.37	0.30	0.66
2dmpa01	99/77	0.26	0.69	0.49	0.48	0.75
2e3tb03	181/129	0.23	0.57	0.36	0.34	0.62
2j8ga03	72/95	0.30	0.71	0.47	0.57	0.68
2jrra01	142/101	0.25	0.65	0.46	0.45	0.70
2kssa01	64/50	0.20	0.69	0.39	0.48	0.68
2lrda00	116/75	0.22	0.68	0.47	0.48	0.65
2xepa03	88/79	0.11	0.56	0.39	0.36	0.68
2yila02	75/109	0.23	0.64	0.44	0.37	0.72
3g5ca02	79/87	0.30	0.72	0.56	0.48	0.79
3pmqa02	73/69	0.21	0.76	0.46	0.64	0.70
4a0ta01	96/76	0.37	0.72	0.56	0.50	0.79
4dcza00	87/82	0.25	0.64	0.43	0.43	0.70
5b1aj00	90/78	0.07	0.48	0.29	0.25	0.50
5daea00	83/72	0.28	0.77	0.48	0.65	0.70
median over domains		0.23	0.67	0.45	0.46	0.68

Table 4: Comparison of face posets built from heavy atoms and from Cα sites on identical frames (md-CATH).

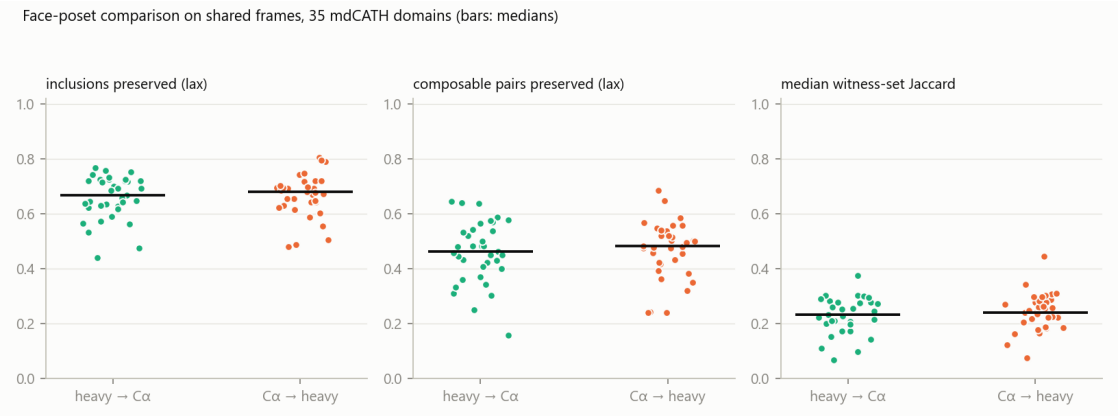


Figure 5: Face-poset comparison between the heavy-atom and Cα representations of each mdCATH domain, in both directions.

6.4 Temperature ladder

temperature (K)	320	348	379	413	450
median β_0	1.0	1.0	1.0	1.0	1.0
median triangles	12	16	19	20	23
median h at τ_{ref} (nm)	0.232	0.287	0.372	0.578	0.861

Table 5: mdCATH temperature ladder over 35 domains (same chart and cover per domain).

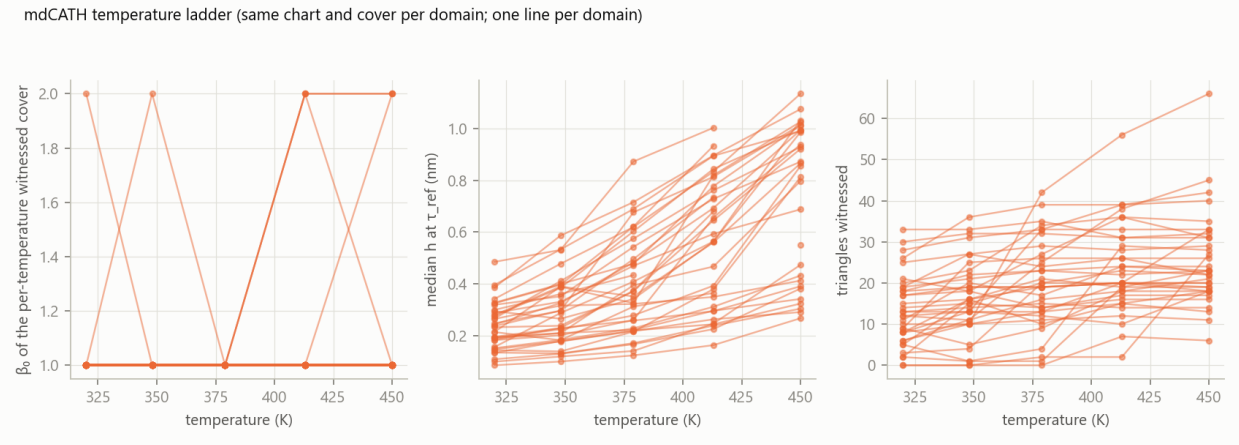


Figure 6: mdCATH temperature ladder: per-temperature witnessed cover and field in the shared chart and cover of each domain.

6.5 Exemplars

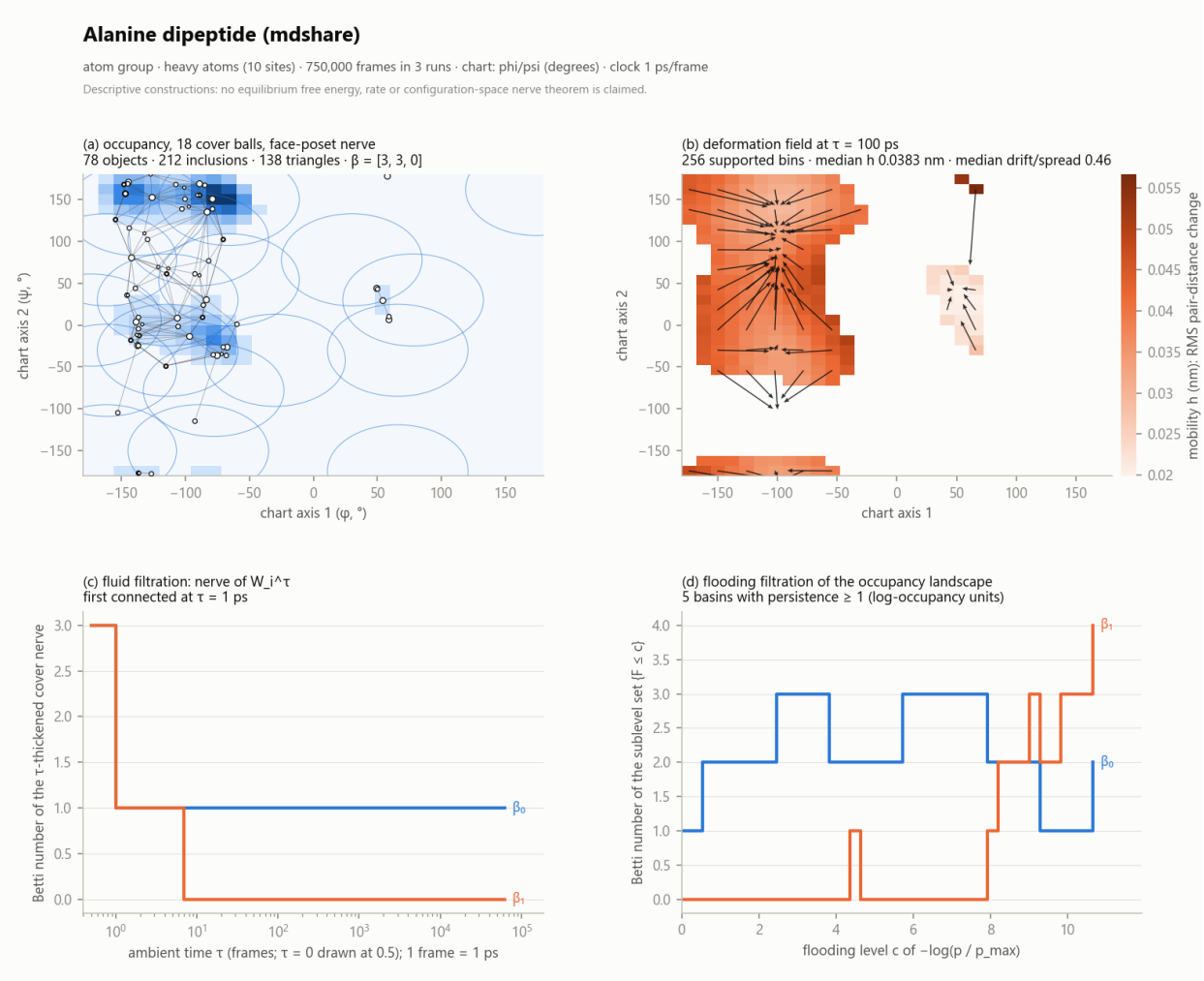


Figure 7: Alanine dipeptide (mdshare): occupancy and nerve, deformation field, fluid and flooding filtrations.

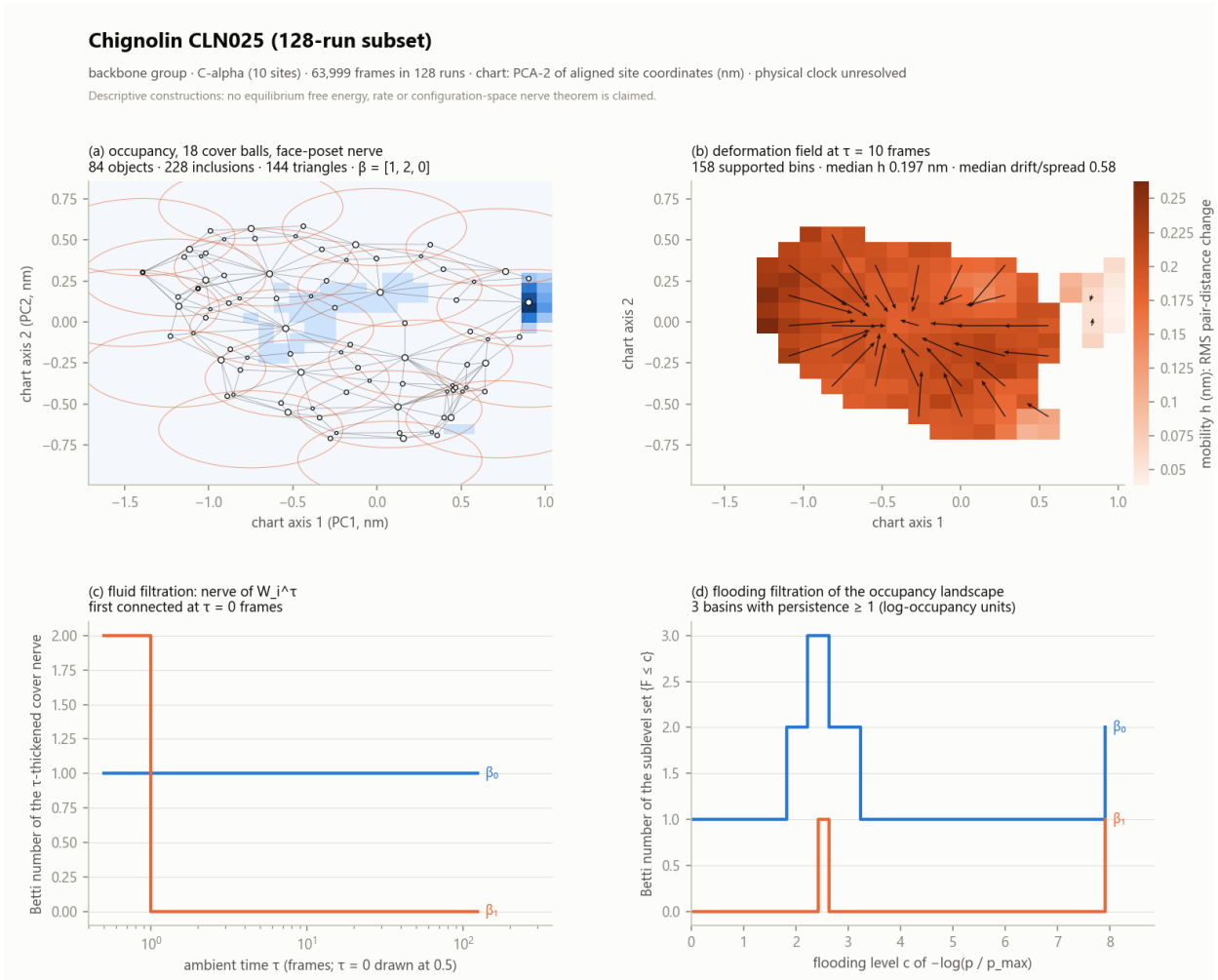


Figure 8: Chignolin CLN025 (128-run subset): occupancy and nerve, deformation field, fluid and flooding filtrations.

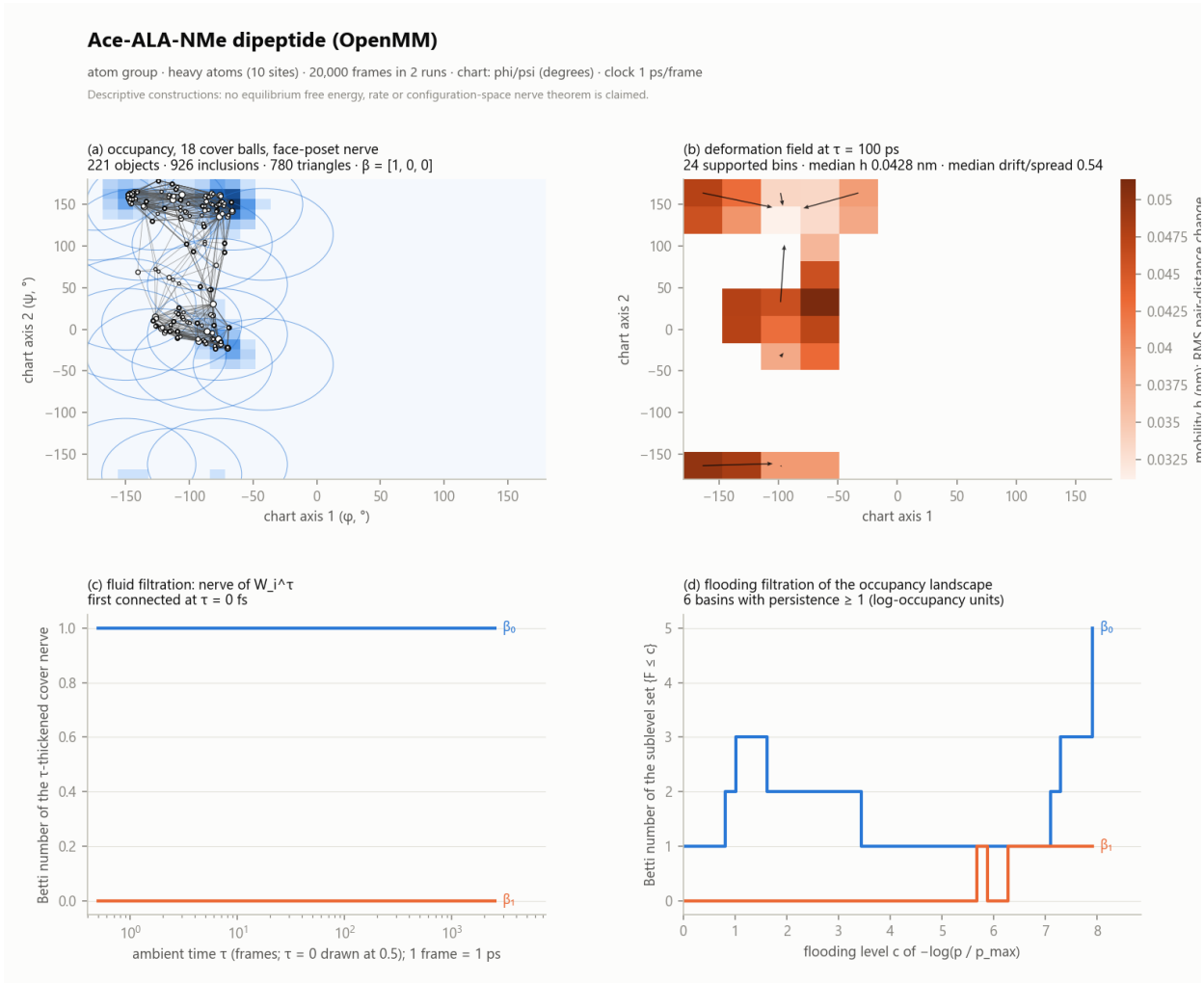


Figure 9: Ace-ALA-NMe dipeptide (OpenMM): occupancy and nerve, deformation field, fluid and flooding filtrations.

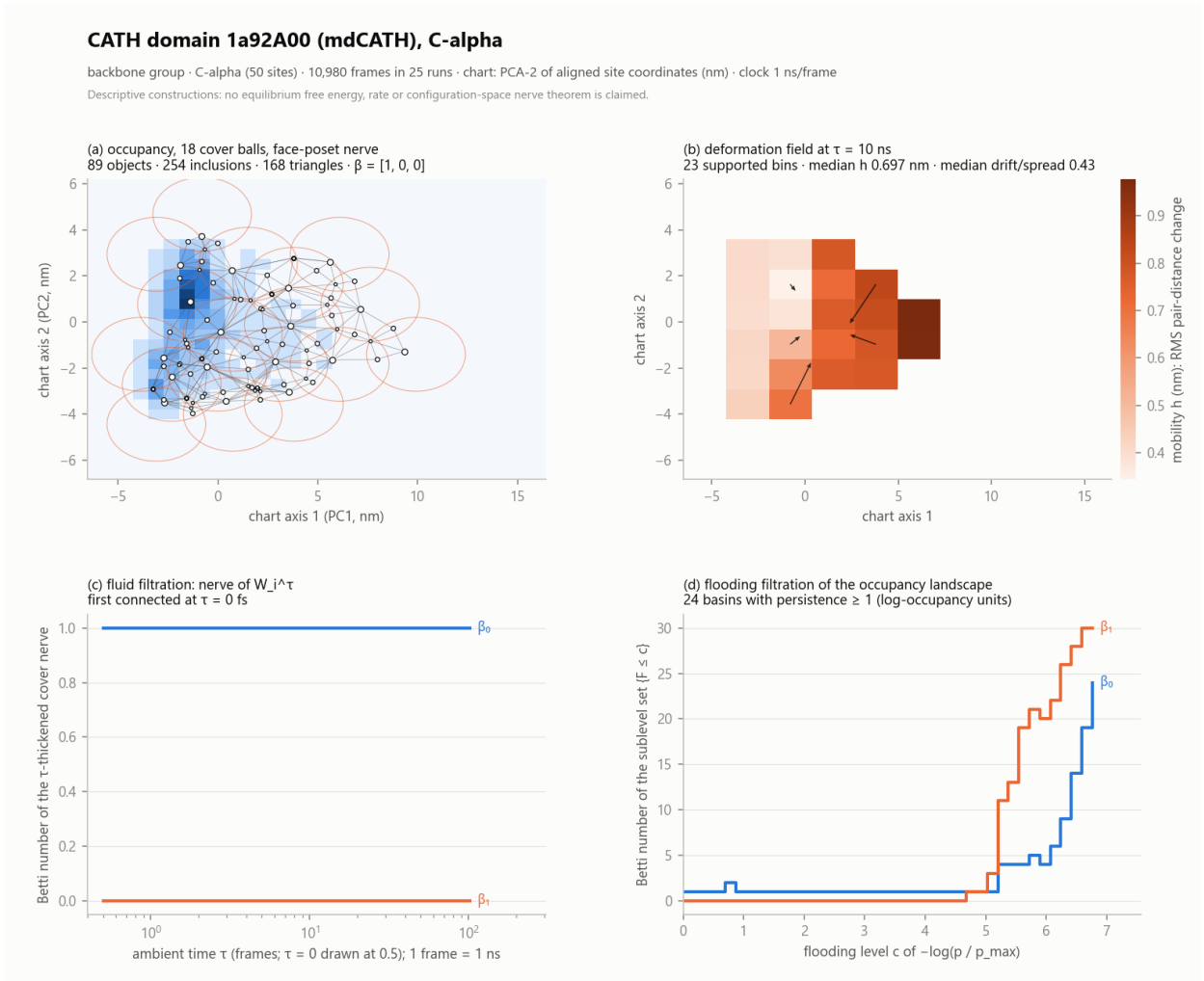


Figure 10: CATH domain 1a92A00 (mdCATH), C-alpha: occupancy and nerve, deformation field, fluid and flooding filtrations.

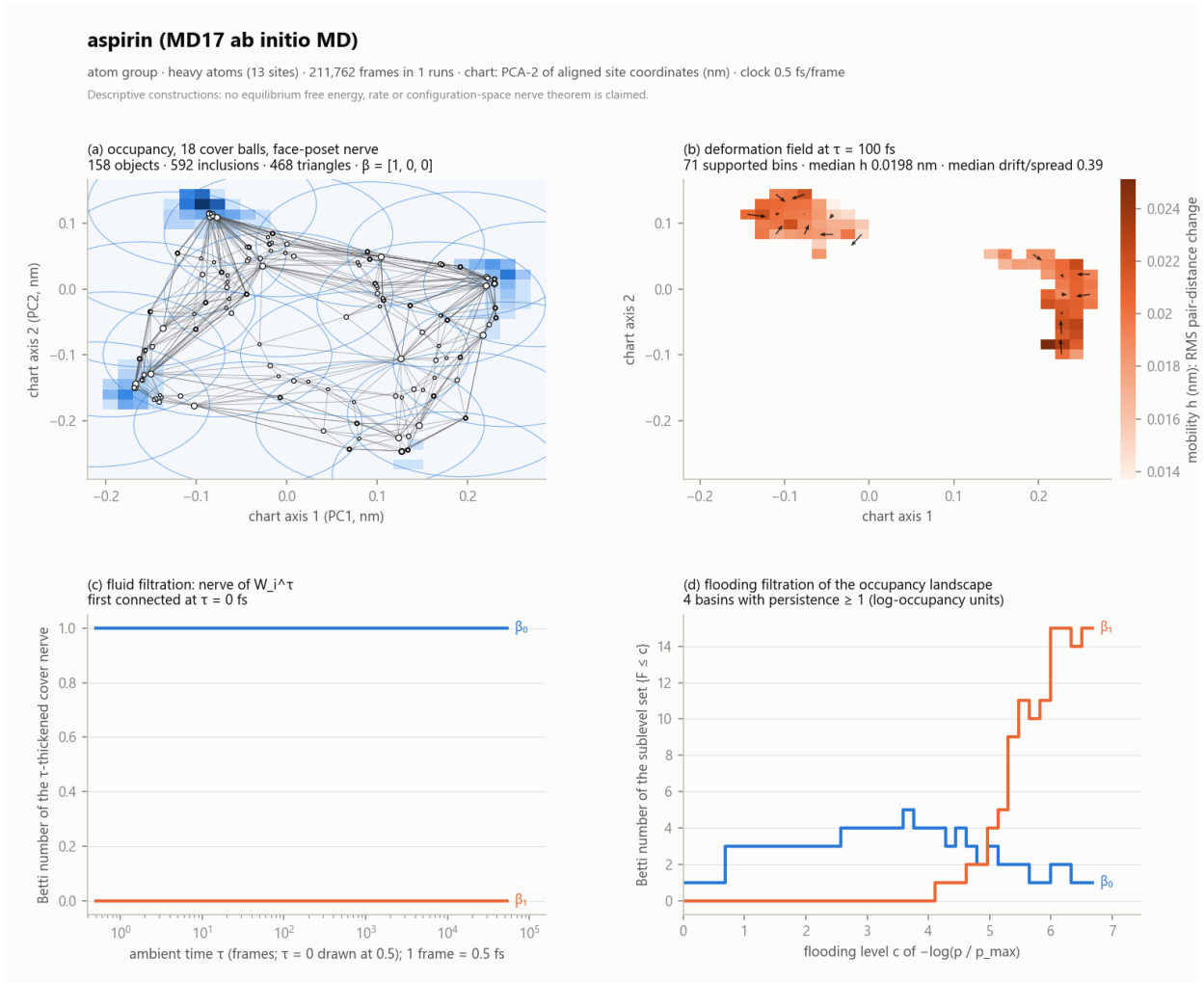


Figure 11: aspirin (MD17 ab initio MD): occupancy and nerve, deformation field, fluid and flooding filtrations.

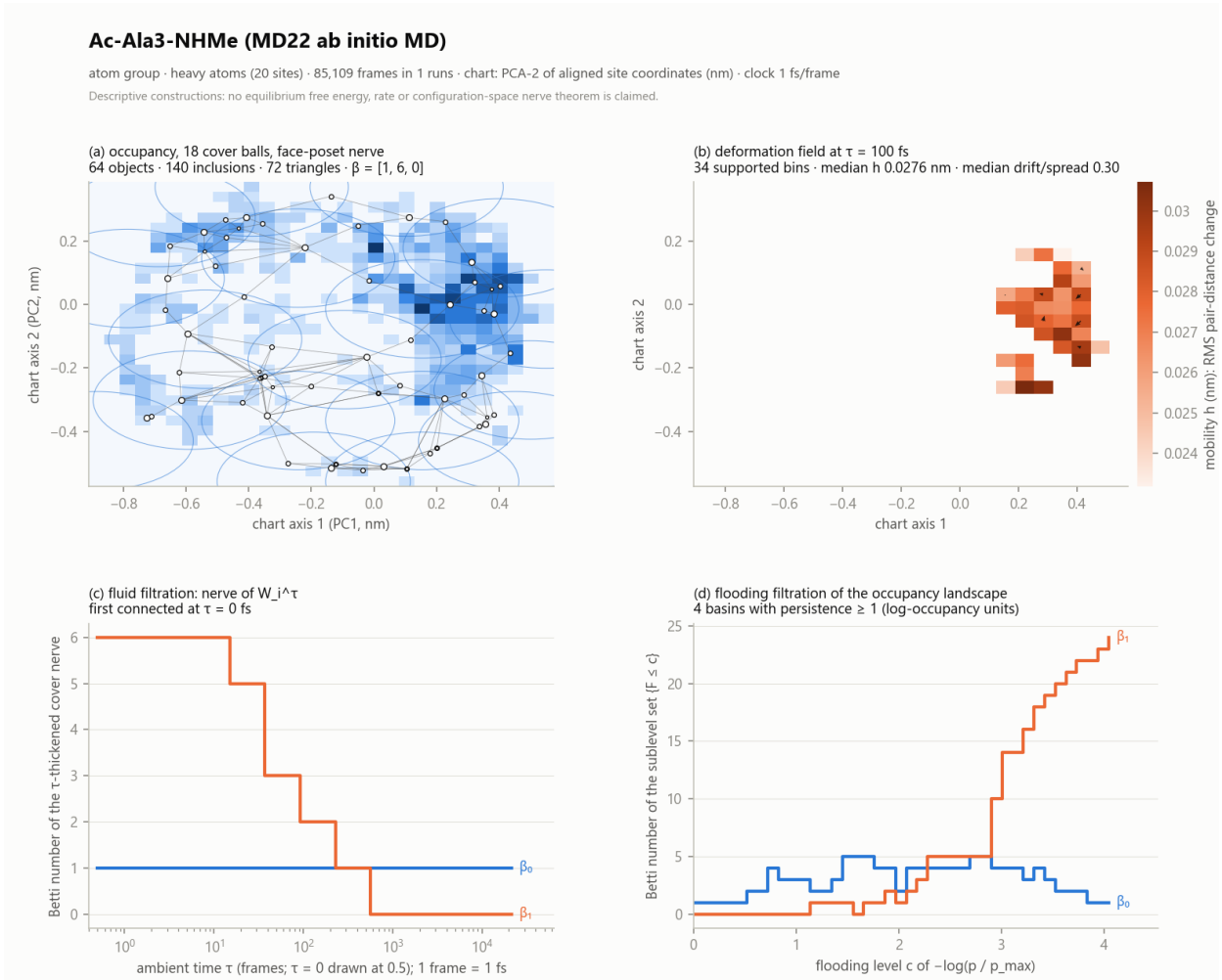


Figure 12: Ac-Ala3-NHMe (MD22 ab initio MD): occupancy and nerve, deformation field, fluid and flooding filtrations.

7 Limitations

Charts are two-dimensional projections (PCA or two dihedrals); the fibre over a chart bin is averaged, and nothing about the full configuration space is asserted. Covers use a single declared radius with a sensitivity ladder, not an optimised scale. Homology is over GF(2). The MD17/MD22 systems are single ab initio trajectories at elevated temperature with nominal time steps taken from the dataset papers; the dipeptide family comes from short new implicit-solvent simulations; mdCATH frames are 1 ns apart, which makes every per-frame quantity saturated; CLN025 has no physical clock. Support thresholds leave bins empty rather than filling them; three single-trajectory MD22 systems (AT-AT, AT-AT-CG-CG, stachyose) revisit chart bins in fewer than five blocks and therefore have no supported mobility at the reference lag. The comparison functor is an empirical best-match map, not a proved functor. The SDG results concern an idealisation (formally open covers, smooth flows) and are used to interpret, not to validate, the measurements. The flooding filtration is a geometric analogue of metadynamics computed from occupancy; no biased simulation was run. None of the constructions is a kinetic model, and the pooled occupancy is not an equilibrium free energy.

8 Reproduction

From `fluid-nerve/`: `python acquire_sgdml.py`, `python acquire_mdcath.py`, `python generate_dipeptides`
`python build_all.py`, `python render.py`, `python export_homotopy.py`, `python compare_all.py`,
`python theory/make_results.py`, `tectonic theory/fluid-nerve-theory.tex`. Unit tests on synthetic data: `python -m unittest discover -s tests`. Raw data are not committed; manifests with SHA-256 hashes are.

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